

\* Electricity - is the set of physical phenomena associated with the presence and motion of matters that has property of electric charge.

2 categories (Based on behaviours)

1. Static
2. Current

Some equations (Maxwell)

Differential form

Integral form

Popular Name

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

Gauss's Law for electricity

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\oint \vec{B} \cdot d\vec{s} = 0$$

Gauss's Law for magnetism

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{\partial \oint \vec{B} \cdot d\vec{s}}{\partial t}$$

Faraday's law of induction

$$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{\partial \oint \vec{E} \cdot d\vec{s}}{\partial t} + \mu_0 \oint \vec{J} \cdot d\vec{s}$$

Ampere's Law (extended)

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\oint \vec{J} \cdot d\vec{s} = -\frac{\partial \oint \rho \cdot dV}{\partial t}$$

Continuity equation

Prob 2

i. Number of electron =  $\frac{1 \text{ C}}{1.602 \times 10^{-19} \text{ C}} \approx 6.25 \times 10^{18}$

ii. a) Charge =  $1.24 \times 10^{18} \times (-1.602 \times 10^{-19}) = -198.67 \text{ mC}$

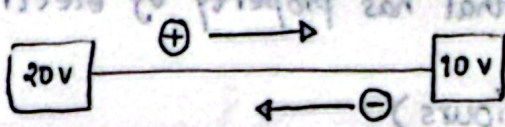
b) Charge =  $-3.941 \text{ C}$

iii.  $W = Vq \Rightarrow V = \frac{W}{q} = \frac{1.2}{20 \times 10^{-6} \text{ C}} = 60 \text{ kV}$



# Voltage/Potential

## Sign Convention



$$W = q(V_{\text{final}} - V_{\text{initial}})$$

+ → for +ve charge  
- → " -ve "

+ → External work done  
- → Work done by charge it-self

## Problem 2

$$i. W = q(V_B - V_A)$$

$$\Rightarrow -10 = (-2)(V_B - 22)$$

$$\Rightarrow V_B = 27 \text{ Volt}$$

$$ii. W = q(V_B - V_A)$$

$$= (-5)(-11 - (-7))$$

$$= 20 \text{ J}$$

## Problem 3

$$W = q(V_B - V_A)$$

$$= (-6 \times 10^{-3})(-30 - 20)$$

$$= 0.3 \text{ J}$$

∴ No the work is done by external force



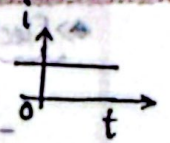
## Current (i)

The change of charge with respect to time

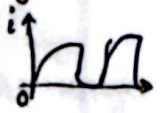
$$i = \frac{\Delta q}{\Delta t} \quad i = \lim_{\Delta t \rightarrow 0} \frac{\Delta q}{\Delta t}$$

$$\therefore i = \frac{dq}{dt} \quad \boxed{i = \frac{q}{t}} \text{ (Ampere)}$$

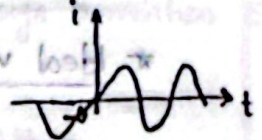
Types: 1. DC



\* Pulsating DC



2. AC



## Power (P)

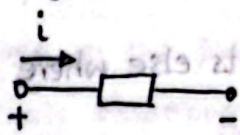
Change of energy with time

$$P = \frac{dW}{dt} \quad \boxed{P = \frac{W}{t}}$$

(Js<sup>-1</sup>) = Watt (W)

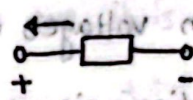
$$P = \frac{dW}{dq} \cdot \frac{dq}{dt} = Vi \quad \boxed{P = Vi}$$

Passive sign convention:



$P = +ve$

Absorbing



$P = -ve$

Supplying

$\Sigma P = 0$  [The energy conservation Law]

## Circuit elements

Active element (and power)  
or 3rd (+ve power)

- Capable of generating energy.
- Is said to be active if it can add some gain (voltage or current) to a circuit.
- Can absorb energy if they are forced to do so by other active element.
- Example: voltage/current sources, generators, transistors, operational amplifiers.

Passive element (+ve power)

- Cannot supply energy. Can only consume/dissipate/store energy.
- Examples: Resistors, capacitors, inductors, transformers.
- Transformers change the voltage or current levels, but the power is unchanged. That's why they are passive element.



## Electrical Sources

- A device that can convert nonelectric energy to electric energy and vice versa.
- Can either deliver or absorb electric power. (Maintain voltage or current)

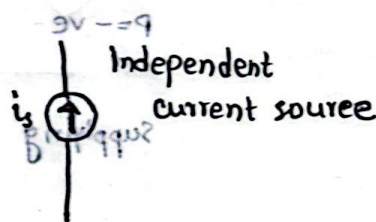
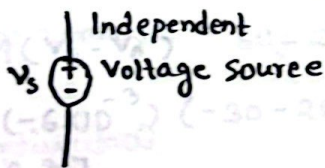
\* Ideal voltage sources maintain a set voltage across their terminals regardless of current flow.

\* Ideal current sources maintain a defined current via their terminals regardless of voltage.

- Can be further described as either independent sources or dependent sources.

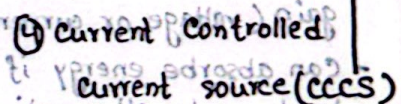
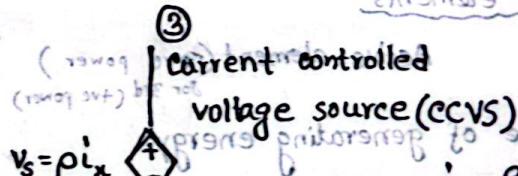
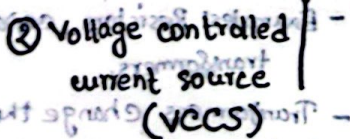
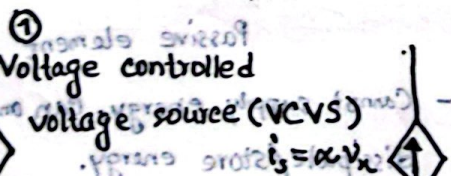
### Independent source

- Does not depend on voltages or currents else where in the circuit.
- Example: power supplies, signal generators and microphones.



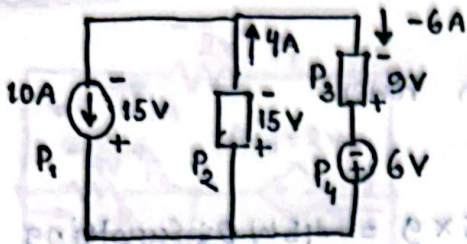
### Dependent / controlled source

- 4 types





**Ex 1**



Find Power of each element.

**Elem 1:**

Current is leaving the + terminal of the voltage. So according to the passive sign convention E1 is supplying power.

**Elem 2:**

It is entering + terminal. E2 is absorbing power.

$$P_2 = +15 \times 4 = +60 \text{ W}$$

**Elem 3:**

Leaving + terminal. E3 is supplying power.

$$P_3 = -9 \times (-6) = +54 \text{ W}$$

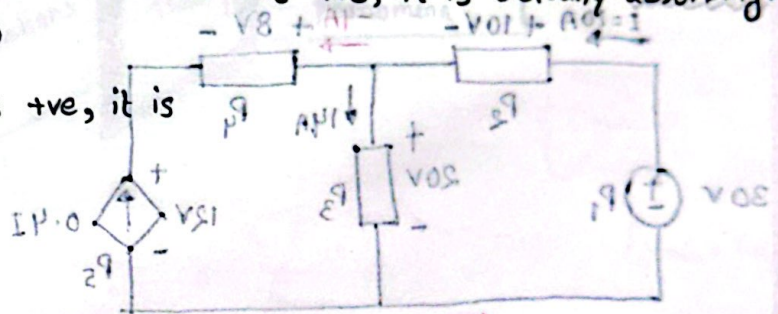
**Elem 4:**

Current leaving + terminal. E4 is supplying.

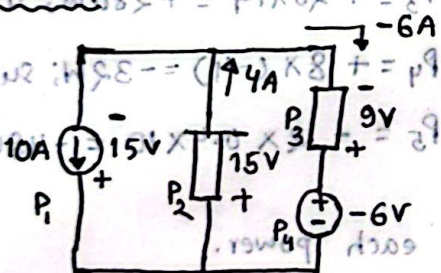
$$P_4 = -6 \times (-6) = +36 \text{ W}$$

However, as the final calc is +ve, it is actually absorbing.

However, as the final form of calculation is +ve, it is actually absorbing.



**Problem 4**



Find each Power.

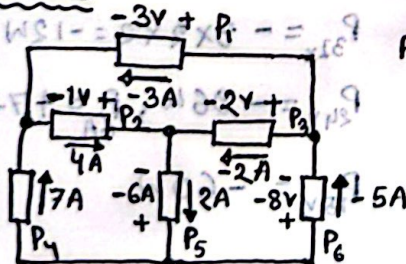
$$P_1 = -15 \times 10 = -150 \text{ W; Supplying}$$

$$P_2 = +15 \times 4 = +60 \text{ W; Absorbing}$$

$$P_3 = -9 \times (-6) = +54 \text{ W; Absorbing}$$

$$P_4 = +(-6) \times (-6) = +36 \text{ W; Absorbing}$$

**Problem 5**



Find each Power.

$$P_1 = +(-3) \times (-3) = -9 \text{ W; Supplying}$$

$$P_2 = -1 \times 4 = -4 \text{ W; Supplying}$$

$$P_3 = +2 \times (-2) = -4 \text{ W; Supplying}$$

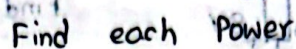
$$P_4 = -5 \times 7 = -35 \text{ W; Supplying}$$

$$P_5 = -2 \times (-6) = +12 \text{ W; Absorbing}$$

$$P_6 = +(-8) \times (-5) = +40 \text{ W; Absorbing}$$



### Problem 6



$P_2 = 2 \times 9 = +18 \text{ W}$ ; absorbing

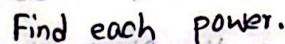
$$P_4 = -3 \times (-5) = +15 \text{ W ; absorbing}$$

$$= +12 \text{ W}; \text{ absorbing}$$

### Verifying Energy conservation Law,

$$\Sigma P = -45 + 18 + 12 + 15 = 0 \quad \text{(Verified)}$$

### Problem 7



$$P_1 = -30 \times 10 = -300 \text{ W; supplying}$$

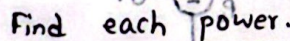
$$P_2 = +10 \times 10 = +100 \text{ W}; \text{ absorbing}$$

$$P_3 = + 20 \times 14 = + 280 \text{ W; absorbing.}$$

$$P_4 = + 8 \times (-4) = -32 \text{ W; supplying}$$

$$P_5 = -12 \times 0.4 \times 10 = -48 \text{ W; supplying}$$

### Problem 8



$$P_1 = +12 \times 2 = +24 \text{ W}, P_2 = -24 \text{ W}$$

$$P_3 = +12W, P_4 = +36W, P_5 = +48W$$

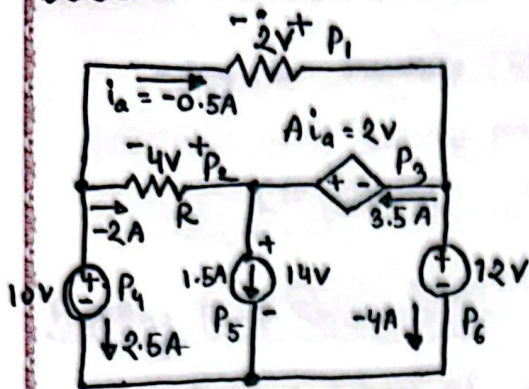
$$P_{22} = -3 \times 2 \times 2 = -12 \text{ W}$$

$$P_{21V} = -96W, P_{21} = 4.72W$$

$$P_{15V} = -60W$$



### Problem 9



•  $R = ?$  gain  $A = ?$  of the ccvs

• Find each Power.

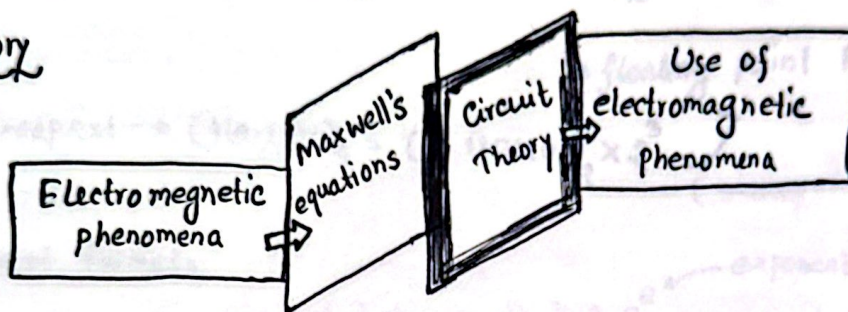
$$\Rightarrow R = \frac{-4}{-2} = 2\Omega, A = \frac{2}{i_a} = \frac{2}{-0.5} = -4V/A$$

$$P_1 = -2 \times (-0.5) = +1W, P_2 = -4 \times (-2) = +8W$$

$$P_3 = -2 \times 3.5 = -7W, P_4 = +10 \times 2.5 = +25W$$

$$P_5 = +14 \times 1.5 = +21W, P_6 = 12 \times (-4) = -48W$$

### Circuit Theory



$$F = \pm (0.1d_1d_2d_3 \dots d_n)_p \times p^e$$

where,  $d_1 = 1$  [For other bases (not binary)]

Convention 2: (IEEE Normalize form)

$$F = \pm (0.1d_1d_2d_3 \dots d_n)_p \times p^e$$

Convention 3: (IEEE Denormalize form)

$$F = \pm (1.d_1d_2d_3 \dots d_n)_p \times p^e$$